22.12. EXERCISES

22.1. What is the purpose of the clay buffer of a waste repository in granitic rock?

22.2. Which of the following ions move slower than the groundwater: K^+, Cs^+, La^{3+}, TcO_4^-, HCO_3^-?

22.3. Why is Np assumed to move faster than Pu in most groundwaters?

22.4. (a) What types of geologic formations are considered for waste repositories? (b) Which types were used/planned for Asse and WIPP?

22.5. What will be the concentration of $Am(CO_3)_2^-$ at pH 7 under the conditions given in §22.7.

22.6. What will be the ratio of [Pu(III)]/Pu[(IV)] in a groundwater containing small concentrations of iron in the relation Fe(II) 99% and Fe(III) 1%? E^0 for Fe(II)/Fe(III) = 0.743 V. Neglect hydrolysis.

22.7. (a) Calculate the Pourbaix-line for Np(IV)/Np(V) in Figure 22.6. (b) What value must the *E*h value of water exceed at pH 6 for Np(V) to dominate? Neglect hydrolysis.

22.8. A miner has a deep well 160 km away from a waste repository. During an earthquake, the rock fractures and a groundwater stream opens between the repository and the well so that 10 Ci ^{90}Sr momentarily dissolves and moves toward the well at ν_w 160 km y^{-1}. (a) What will be the amount of ^{90}Sr (in Bq) reaching the well (assume plug flow)? (b) Will that water be harmful to the miner?

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APPENDIX A

Solvent Extraction Separations

Contents

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Solvent extraction is a technique which has been highly developed within many national nuclear energy programs because of its suitability as a selective separation process for fission products, actinides, and other radioactive substances. The technique is briefly described in several sections of this book (Sections 14.2.4.3, 14.3.4, 17.7.3.4, 18.2.6, 18.4.3, 21.7, and especially in 21.6 on the Purex process). It is based on the formation of uncharged organic metal complexes which are preferentially soluble in organic solvents. The three main types of such compounds are:

1. Organic chelate (i.e. binding through two positions in the organic molecule, like a claw) complexes such as plutonium tetraacetylacetonate, $PuAa_4$ (general formula MA_n) (Section 18.4.3, cf. Figs 18.3, 18.9 and 18.11);
2. Inorganic metal complexes forming adducts with solvating organic compounds like TBP and hexone, e.g. $UO_2(NO_3)_2)(TBP)_2$ (or generally ML_nS_s, where L is an inorganic anion and S is the organic adduct former (Section 21.6, cf. Fig. 14.9);
3. Ion pair complexes between large organic cations (like alkylammonium ions, R_3NH^+) and negatively charge inorganic complexes like $UO_2(SO_4)^-$ (generally RB^+ML^-, Section 14.2.4.2).

The physical and chemical principles governing the formation of extractable metal complexes, the conclusions which can be drawn about the chemical system from solvent extraction studies, and analytical and industrial applications are described in several monographs, e.g. 'Principles and Practices of Solvent Extraction' (2004).

A.1. SINGLE STAGE BATCH EXTRACTIONS

Let us consider a solute (e.g. uranium) which is distributed between an organic and an aqueous phase, independent of the kind of compounds the solute forms in the two